DATA SCIENCE & ANALYTICS

MODULE 04 SESSION 17

Power Query II: files that arrive as files

Your first report, then Append. Folder, store-list cleaning and Merge follow next lecture.

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Sun 1 Nov 2026

ACROSS THE FILE-PREPARATION LESSONS

The Power Query sequence

01 Append queries, and say why names matter

02 Combine a folder of files into one table

03 Split columns and resolve error cells

04 Build a clean query as a Reference

05 Merge with the right join kind

EVERYONE DO

Step 1.1

Review Modules 1–3

Nakheel data, the SQL you wrote, and choosing a model for a business question. Then your first Power BI report.

INSTRUCTOR DO

Step 1.2

How much did the shops sell in each governorate?

Two yearly files from finance, twelve files from the shops, a store list typed by hand.

shared/datasets/nakheel_stores/nakheel_store_files.zip

D17 DOWNLOAD · SEPARATE FROM D16 · shared/datasets/nakheel_stores

Get the store files before you start

Inside nakheel_store_files	Check
finance/store_sales_2024-25.csv	892 orders
finance/store_sales_2025-26.csv	1,009 orders
store_files/	12 CSVs and notes.txt
stores_raw.csv	12 stores

Save: D17-checkpoint.pbix (Desktop), or report D17-checkpoint on model D17-nakheel (browser). Keep one model throughout.

YOU KNOW THESE FROM SQL · TODAY IN POWER QUERY

The same ideas, a new place

SQL you wrote	Power Query today	Where
UNION ALL	Append queries	activity 01
many tables stacked	a folder of files, Source.Name	activity 02
SPLIT, SAFE_CAST	Split Column, Replace Errors	activities 04–05
a view on a table	a Reference query	activity 04
LEFT JOIN, and … IS NULL	Merge: Left Outer, Left Anti	activities 06–07

Nakheel SQL check: completed store orders total 205,338.00 dinars.

A query is one table and the steps that make it, listed by name in the Queries pane on the left of Power Query. Append, Reference and Merge each build a query from other queries.

PQ · 01

Two years in one table

Finance sends store sales once a year. Stack the two files, and find out why half the revenue is missing.

1 · store_sales	2 · store_files_sales	3 · stores	4 · store_sales_by_store
Append	Folder	Reference, Split	Merge

DEFINITION · HOME → COMBINE → APPEND QUERIES

Append puts the rows of one query under the rows of another, in one table: the Power Query version of UNION ALL. Choose Append queries as new, so the two inputs stay as they are.

INSTRUCTOR DO

Step 2.1

Append is UNION ALL, matched by name

Two yearly files, one renamed column, and a card that reads less than half.

power-query/labs/D17/01-Evr-Append-Two-Years/README.md

BEFORE → APPEND QUERIES AS NEW → AFTER · THE TWO FINANCE FILES

Append lines columns up by name

store_sales_2024-25.csv · 892 rows

order_id	store_code	order_date	status	order_total
100001	NK-KRK-01	2024-09-01	completed	22.50

store_sales_2025-26.csv · 1,009 rows

order_id	order_date	store_code	status	amount
101340	2025-09-01	NK-AMM-03	completed	30.00

indigo: the same names, in another order · **amber**: a new name

store_sales · 1,901 rows · 6 columns

order_id	store_code	order_date	status	order_total	amount
100001	NK-KRK-01	2024-09-01	completed	22.50	*null*
100003	NK-AMM-01	2024-09-03	completed	47.00	*null*
101340	NK-AMM-03	2025-09-01	completed	*null*	30.00
101341	NK-AMM-03	2025-09-01	cancelled	*null*	25.00

SQL UNION ALL pairs by position: this date meets store_code; incompatible types may reject it. **2025-09-01**

By name: the swap did no harm; the new name became a sixth column (amber).

WHAT COMES BACK · A CARD ON SUM OF order_total · COMPLETED STORE ORDERS

Half the revenue sits in a column the card does not add

Completed store revenue in SQL: **205,338.00**

order_total, 2024-25	97,743.00	amount, 2025-26	107,595.00

`Append as it came`	the card says 97,743.00: only the first year	×
`Combine the amount columns later`	a saved step also replays, but leaves the inputs with different names	✓
`Rename amount at the source`	the card says 205,338.00, at every refresh	✓

Rename amount in store_sales_2025-26: every dependent query then reads one order_total column.

EVERYONE DO

Step 2.2

Append two years, and keep the inputs out of the model

Activity 01: append, rename amount, then save D17-checkpoint. Turn off input loading if your editor offers it.

power-query/labs/D17/01-Evr-Append-Two-Years/README.md

ENABLE LOAD (RIGHT-CLICK A QUERY): IS ITS RESULT LOADED INTO THE MODEL AS A TABLE OF ITS OWN?

Rename at the source, load only the result

Query	What you do	In the model?
store_sales_2025-26	rename amount → order_total	no: untick Enable load
store_sales_2024-25	nothing	no: untick Enable load
store_sales	nothing: it reads both again	**yes: 1,901 rows, 5 columns**

Completed card: 205,338.00. Inputs still run at refresh. No Enable load in your browser? Keep them and report from store_sales only.

THE RULE TO KEEP

Fix the input, not the result.

A fix in the query that reads the file reaches every query built on it, at every refresh.

PQ · 02

Twelve shop files at once

Each shop sends one file named after itself. Read the folder, not the files.

1 · store_sales	**2 · store_files_sales**	3 · stores	4 · store_sales_by_store
Append	Folder	Reference, Split	Merge

INSTRUCTOR DO

Step 3.1

A folder of files

Look at the file list, keep the CSV files, combine, and turn the file name into a store code.

power-query/labs/D17/02-Evr-Combine-A-Folder/README.md

One folder, thirteen files, one table

the folder · one row is one file

Content	Name	Extension
Binary	NK-AMM-01.csv	.csv
Binary	NK-AMM-02.csv	.csv
Binary	NK-AMM-03.csv	.csv
Binary	NK-AQB-01.csv	.csv
Binary	NK-IRB-01.csv	.csv
Binary	NK-IRB-02.csv	.csv
Binary	NK-KRK-01.csv	.csv
Binary	NK-MDB-01.csv	.csv
Binary	NK-MFQ-01.csv	.csv
Binary	**NK-POP-01.csv**	.csv
Binary	NK-SLT-01.csv	.csv
Binary	NK-ZRQ-01.csv	.csv
Binary	**notes.txt**	**.txt**



keep .csv
Combine Files

store_files_sales · 1,901 rows

store_code	order_id	order_date	order_total
NK-AMM-01	100003	2024-09-03	47.00
NK-AMM-02	100006	2024-09-03	78.50
NK-AMM-03	100004	2024-09-03	120.00
NK-AQB-01	100016	2024-09-07	80.00

amber: notes.txt is not a sales file: filter it out first. **lime**: Source.Name, the file name, becomes store_code (Replace Values: .csv with nothing). **indigo**: NK-POP-01.csv, a code to remember.

Take NK-MFQ-01.csv out, Refresh: **1,855** rows. Put it back: **1,901**. Helper queries apply the sample's steps to every file.

Check Abdali: 251 rows. Clear that filter: 1,901 rows. Completed card: 205,338.00.

EVERYONE DO

Step 3.2

Combine the twelve shop files

Activity 02, steps 1–5. Browser: test the OneDrive/SharePoint folder route; use Desktop or pair if access fails.

power-query/labs/D17/02-Evr-Combine-A-Folder/README.md

STUDENT DO

Which shop, which month?

Three questions a regional manager would ask, from the table you just combined.

power-query/labs/D17/03-Stu-Revenue-By-Store-And-Month/README.md

STUDENT DO

The top shop, the best month, the odd file

INSTRUCTIONS	From store_files_sales: 1. Highest completed revenue by shop? 2. Best and worst calendar month? 3. Which code is the stall? Count its orders across all statuses and read notes.txt.
HINT	Line chart: keep Year and Month, then expand all down one level.
BONUS	A store_code slicer: do the Irbid shops peak together?

Full steps and Check yourself: power-query/labs/D17/03-Stu-Revenue-By-Store-And-Month/README.md

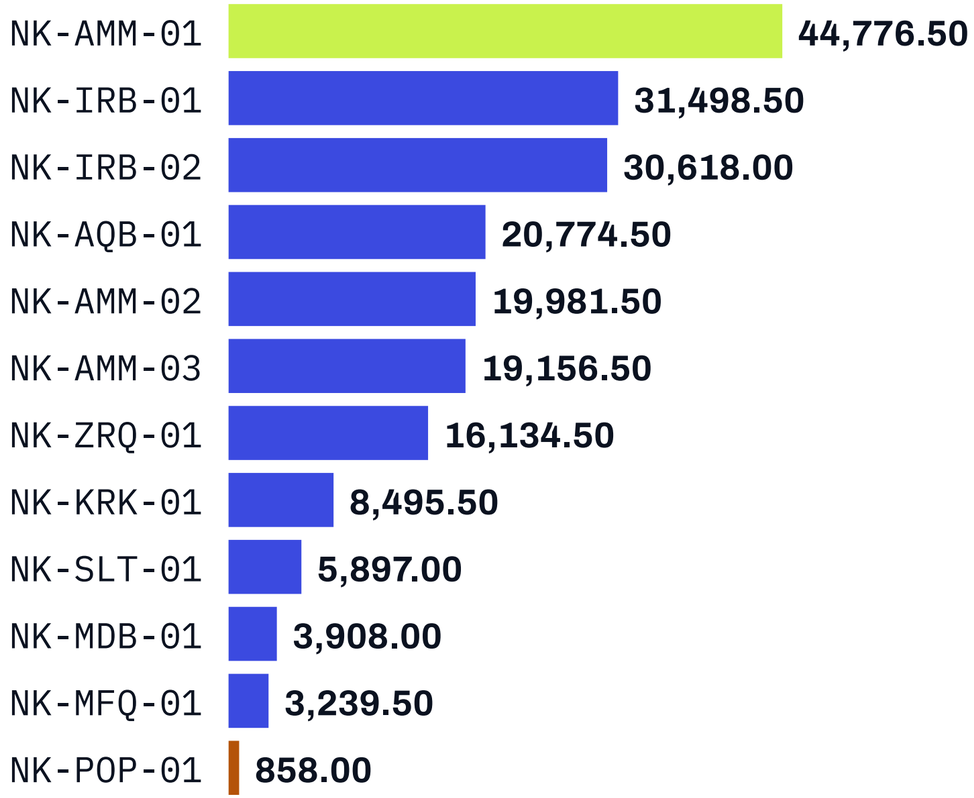
REVIEW

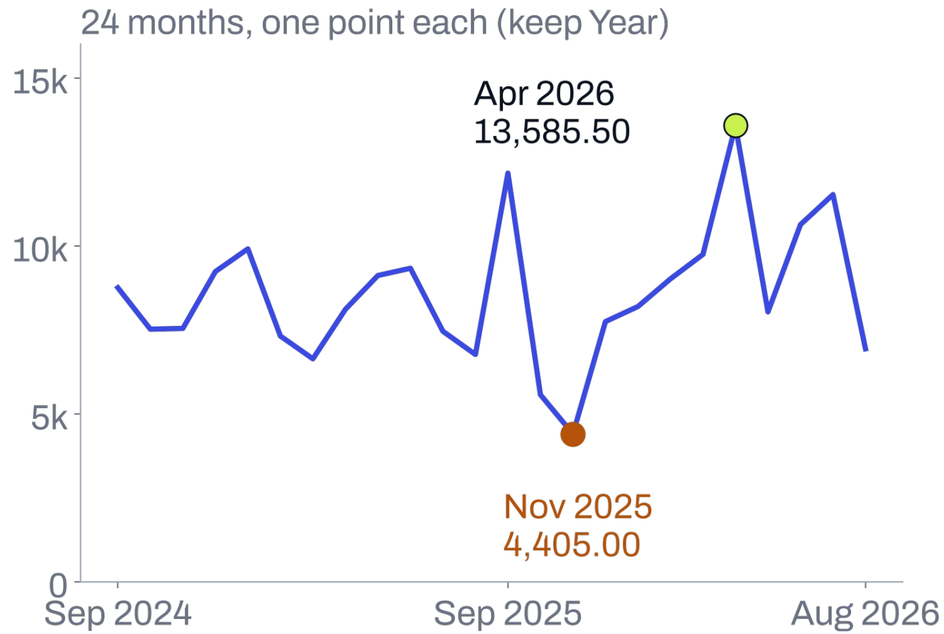
Time's up.
Let's review.

Which shop, which month?

`Solved: power-query/labs/D17/03-Stu-Revenue-By-Store-And-Month/Solved/solution.md`

Abdali (NK-AMM-01) leads; one code is a stall





The twelve bars add up to 205,338.00. NK-POP-01 (amber) is the Ramadan stall: 19 orders, 18 of them completed, 15–28 March 2025. After the break it finds no store.

THE RULE TO KEEP

The folder is the query.

Add a file with the same columns and new orders, then refresh. Overlapping exports would count the same orders twice.

PQ · 03

A store list typed by hand

Three facts in one column, sizes as text, one size unknown. Clean it in a query that reads the raw one.

1 · store_sales	2 · store_files_sales	**3 · stores**	4 · store_sales_by_store
Append	Folder	Reference, Split	Merge

DEFINITION · RIGHT-CLICK A QUERY → REFERENCE

A Reference is a new query that starts from another query’s result. Change the first, and the reference follows; a Duplicate copies the steps and then changes independently.

INSTRUCTOR DO

Step 4.1

Reference, not Duplicate

Keep stores_raw exactly as the file; clean in stores, a reference to it.

power-query/labs/D17/04-Evr-Split-Reference-Errors/README.md

TWO WAYS TO START A SECOND QUERY · THEIR APPLIED STEPS

Reference reads; Duplicate copies

stores · right-click stores_raw → Reference

Source = stores_raw

stores_raw (2) · right-click → Duplicate

Source = Csv.Document(…stores_raw.csv…)

Promoted Headers

Changed Type

Add a step in stores_raw: rename store_name to name. Which inherits it?

Reference	inherits the new rename step: the column is now name	✓
Duplicate	keeps its copied steps: the column is still store_name	×

Both still reread the CSV at refresh. Duplicate copies steps, not a frozen table. Delete the temporary rename step before continuing.

EVERYONE DO

Step 4.2

Split the columns, and an error that is a fact

Activity 04: three facts into three columns, a size into a number, Jerash’s unknown size.

power-query/labs/D17/04-Evr-Split-Reference-Errors/README.md

WHAT COMES BACK · SPLIT AT EACH “ - ”

One column, three facts, three columns

governorate - city - area	gov_code	governorate	area
AM - Amman - Abdali	AM	Amman	Abdali
IR - Irbid - Irbid	IR	Irbid	Irbid
BA - Balqa - Salt	BA	Balqa	Salt
JA - Jerash - Jerash	JA	Jerash	Jerash

The header says city; the values are a code, the governorate and the area. governorate has 9 values. Type the delimiter with its spaces, or every value keeps a space.

SPLIT size AT THE FIRST SPACE: size.1 (THE NUMBER), size.2 (THE UNIT) · size.1 AS WHOLE NUMBER

An Error cell is a message about one value

size (text)	size.1 as Whole Number	Then: Replace Errors → null
420 m²	420	420
380 sqm	380	380
260 m2	260	260
unknown (Jerash)	**Error**	null

The preview still shows the table. Fix the cell before loading the model. A step-level error stops the whole query.

QUICK CHECK

Jerash’s size is unknown. Remove Errors or Replace Errors?

Select size.1. Remove Errors deletes rows with an error there. Replace Errors puts a value of your choice, such as null, in the cell and keeps the row (SQL’s SAFE_CAST gives NULL). Which one keeps the store list true?

THE ANSWER · KEEP ERRORS → LOOK → DECIDE

Replace the error; do not delete the shop

Column quality on size.1 (renamed size_m2 at the end)



Error 8% · Jerash, “unknown”

Is Jerash still in the store list?

Remove Errors	11 rows: Jerash is gone, and after the break no join can find it	×
Replace Errors → null	12 rows: Jerash stays, size empty; known sizes total 2,880 m²	✓

EXAMPLE (lab 10, one shop’s till file): a sale typed as “93.50 JOD” is an error cell. Remove Errors drops it, and completed revenue falls from 2,032.50 to 1,939.00.

STUDENT DO

Finish the store list

Latitude and longitude, a short name for charts, and the facilities manager’s question.

power-query/labs/D17/05-Stu-Clean-The-Store-List/README.md

Column From Examples makes a new column from values you type: type Abdali beside Nakheel Abdali, and Power Query writes the step that does the same for every row.

STUDENT DO

Position, short name, floor area

INSTRUCTIONS	In stores: 1. Split lat / long at “ / ”; use Decimal Number. 2. Column From Examples: Abdali, University Street. Name it short_name; check 12 results. 3. opened: Date. 4. Floor area in Amman and nationally?
HINT	Decimal errors: use English (United States) locale. Browser fallback: Custom Column in the README.
BONUS	Column From Examples on the raw size: what about unknown?

Full steps and Check yourself: power-query/labs/D17/05-Stu-Clean-The-Store-List/README.md

REVIEW

Time's up. Let's review.

The store list

Solved: power-query/labs/D17/05-Stu-Clean-The-Store-List/Solved/solution.md

THE ANSWERS · STORES, 12 ROWS

A list you can filter, sum and map

store_code	short_name	governorate	latitude	longitude	size_m2
NK-AMM-01	Abdali	Amman	31.9632	35.9106	420
NK-IRB-01	University Street	Irbid	32.539	35.856	310
nk-irb-02	Irbid City Centre	Irbid	32.5556	35.85	240
NK-JRS-01	Jerash	Jerash	32.2808	35.8993	null

Floor area: Amman 1,060 m² · all shops 2,880 m² (Jerash unknown). Look again at row 3’s code.

Break

Stretch, then Merge

After the break: the sales meet the store list.

PQ · 04

Sales with their store

Only the store list knows the governorate. Join it to the sales, and read the match count first.

1 · store_sales	2 · store_files_sales	3 · stores	**4 · store_sales_by_store**
Append	Folder	Reference, Split	Merge

DEFINITION · HOME → COMBINE → MERGE QUERIES

Merge finds, for each row of the first query, the rows of the second with the same key, and attaches them as a table you expand: the Power Query version of a JOIN.

INSTRUCTOR DO

Step 5.1

Merge is a JOIN; read the match count

store_sales on the left, stores on the right, by store_code, Left Outer.

power-query/labs/D17/06-Evr-Merge-Sales-And-Stores/README.md

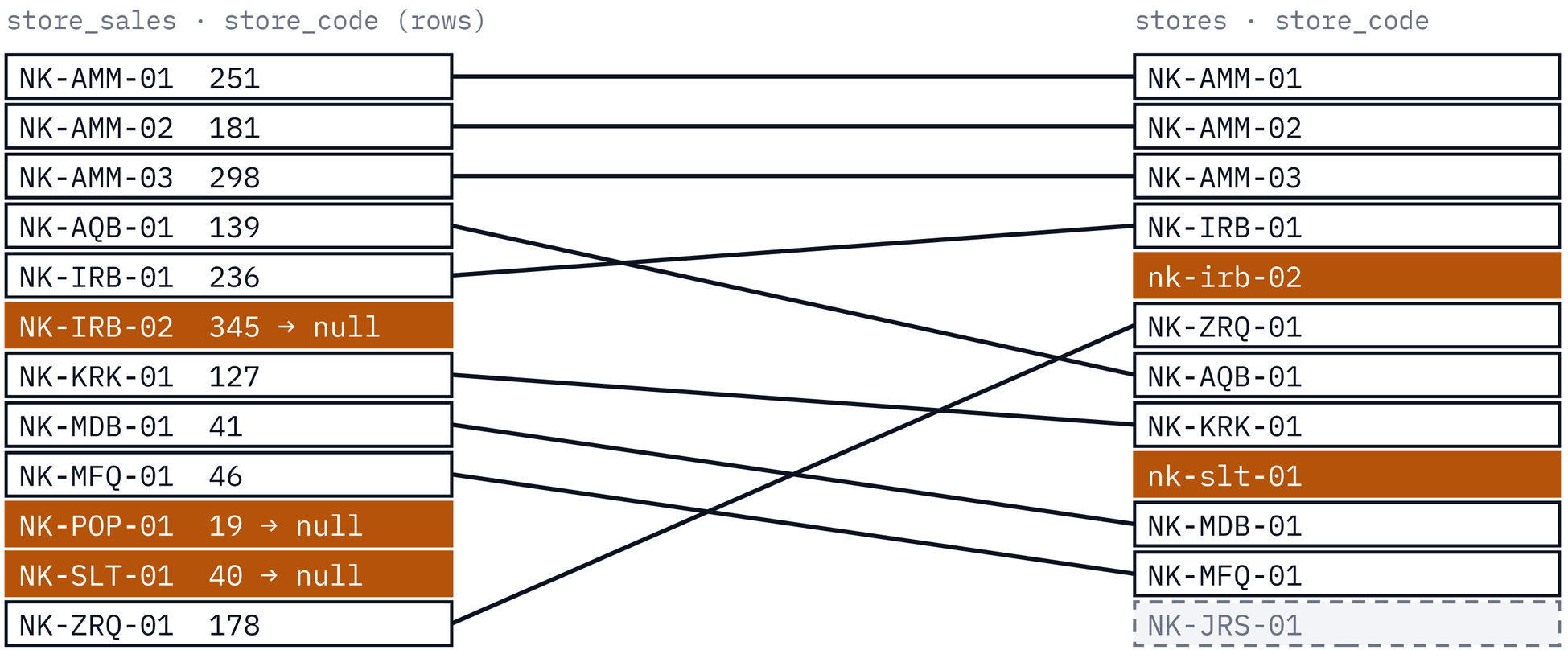
JOIN KIND IN THE MERGE DIALOG · AND THE SQL YOU WROTE

Six join kinds, all familiar

Join kind	Keeps	SQL
Left Outer	every left row, matches attached	LEFT JOIN
Inner	only rows that match	JOIN
Full Outer	every row of both	FULL OUTER JOIN
Right Outer	every right row	RIGHT JOIN
Left Anti	left rows with no match	LEFT JOIN … WHERE right key IS NULL
Right Anti	right rows with no match	the same, the other way

MERGE · LEFT OUTER · EXPECTED FULL-DATA COUNTS, BEFORE THE KEY FIX

Three codes find no store: 404 sales



1,497
of 1,901 match

Amber: no partner. Power Query compares text exactly: items that differ only in case are “different, not identical” (Microsoft). Dashed: Jerash, no sales yet.

AFTER OK · THE NEW COLUMN HOLDS A TABLE IN EVERY ROW

Expand the store’s columns you need

store_sales_by_store · the merge adds a column called stores

order_id	store_code	order_total	stores
100003	NK-AMM-01	47.00	Table
100001	NK-KRK-01	22.50	Table

Each Table is that sale’s matching rows of the store list, one row when the key matches, none when it does not.



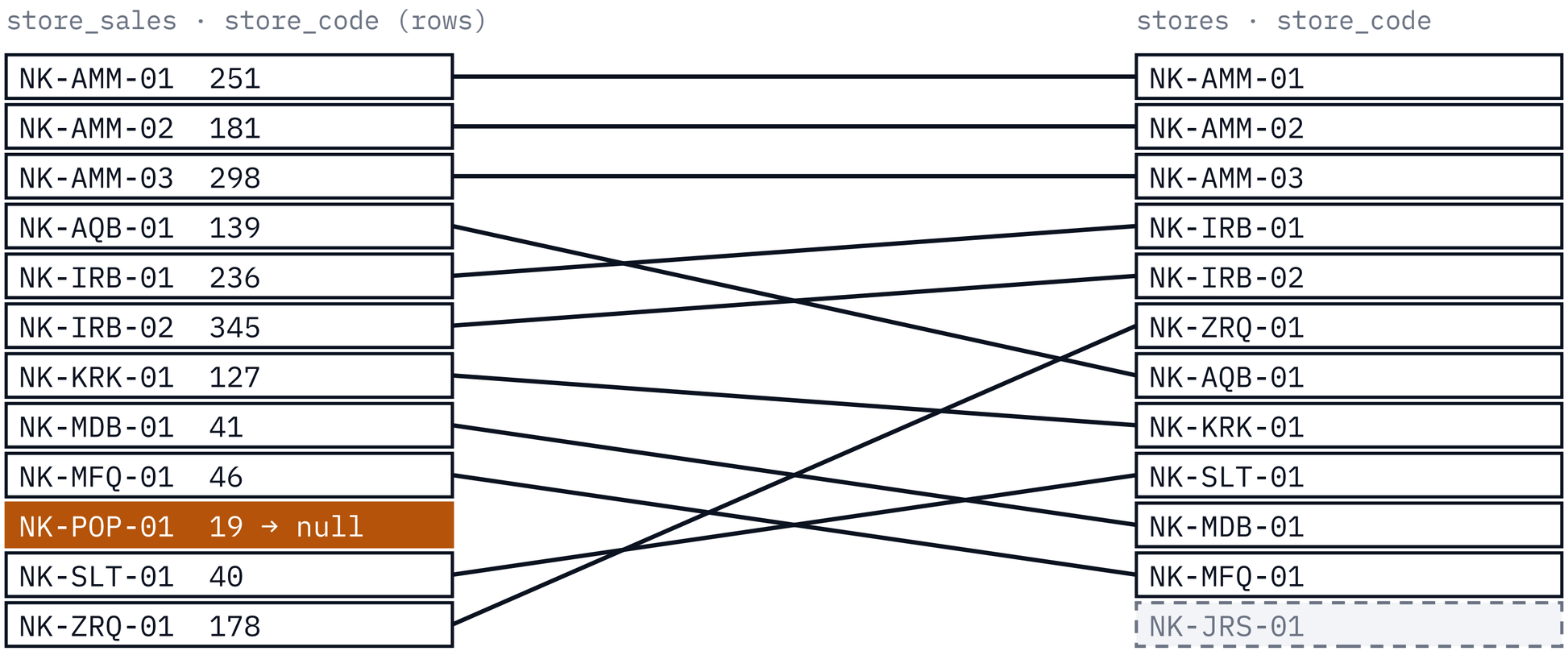
expand governorate, area, short_name · untick “Use original column name as prefix”

order_id	store_code	order_total	governorate	area	short_name
100003	NK-AMM-01	47.00	Amman	Abdali	Abdali
100001	NK-KRK-01	22.50	Karak	Karak	Karak

Filter governorate to null: 404 rows. Delete the temporary Filtered Rows step: all 1,901 return before the key fix.

MERGE · AFTER TRANSFORM → FORMAT → UPPERCASE ON store_code IN stores

One sale code and one shop are left without a partner



1,882
of 1,901 match

Filter null governorates: 19 rows, all NK-POP-01. Delete that temporary filter: all 1,901 return before charting.

ILLUSTRATIVE · IF THE STORE LIST HAD NK-AMM-01 TWICE

A repeated key on the right adds rows

Each of Abdali's 251 sales finds two store rows, so it comes back twice. **The match message does not warn you; the row count does.**

Rows after the merge	2,152, not 1,901	×
Completed revenue	250,114.50, not 205,338.00: Abdali counted twice	×
The check	a key must be unique in the store list: Keep Duplicates on store_code finds it	✓

D06 said the same of SQL joins: a key that repeats on one side multiplies the rows it meets.

EVERYONE DO

Step 5.2

Merge, fix the key, chart by governorate

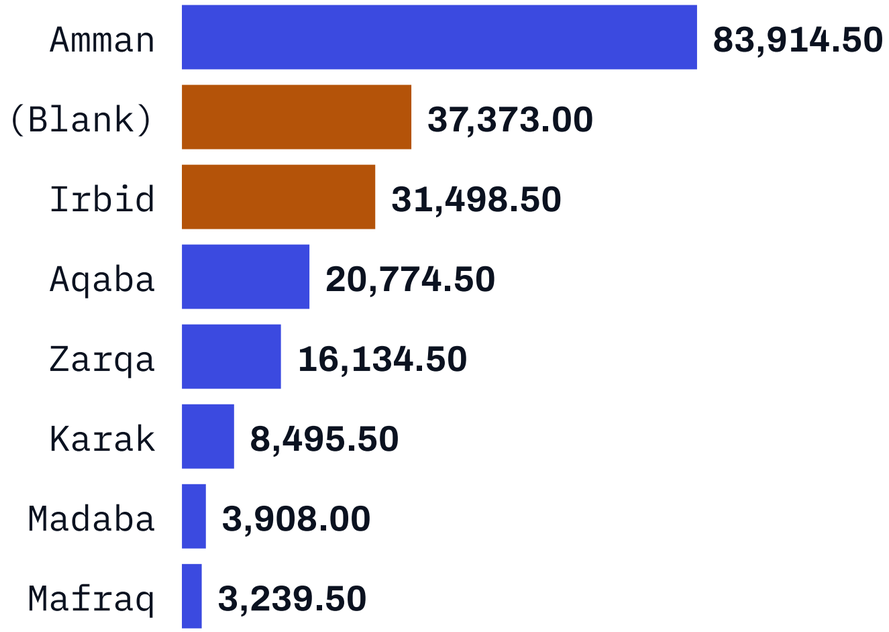
Activity 06: merge, expand, find the 404, UPPERCASE in stores, then a bar chart.

power-query/labs/D17/06-Evr-Merge-Sales-And-Stores/README.md

WHAT COMES BACK · COMPLETED REVENUE BY GOVERNORATE · LEFT OUTER

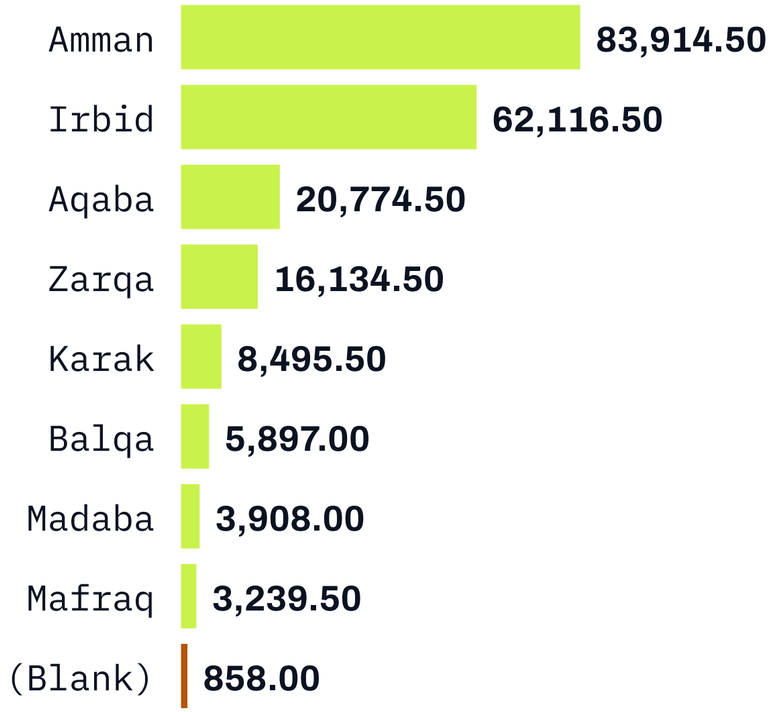
Two lower-case codes moved 36,515.00 into (Blank)

BEFORE · the codes as typed



amber: wrong · Irbid lacks IRB-02 · no Balqa bar

AFTER · UPPERCASE in stores



store_sales_by_store, status = completed: 205,338.00. Clear the query’s null filter. (Blank) keeps the stall’s 858.00.

STUDENT DO

Who has no match?

An auditor’s three questions after any merge: Left Anti, Right Anti and Inner.

power-query/labs/D17/07-Stu-Sales-Without-A-Store/README.md

STUDENT DO

Sales with no store, a store with no sales

INSTRUCTIONS	Start each merge from store_sales. 1. Left Anti: sales without a store? 2. Right Anti: stores without sales? 3. Inner: rows and completed revenue?
HINT	Count all statuses. Revenue visuals: completed only. Left Anti is LEFT JOIN … WHERE right key IS NULL.
BONUS	Turn off audit-query loading if Enable load is available.

Full steps and Check yourself: power-query/labs/D17/07-Stu-Sales-Without-A-Store/README.md

REVIEW

Time's up.
Let's review.

Who has no match?

`Solved: power-query/labs/D17/07-Stu-Sales-Without-A-Store/Solved/solution.md`

Inner loses the stall; Left Outer keeps it as (Blank)

matched to a store 204,480.00

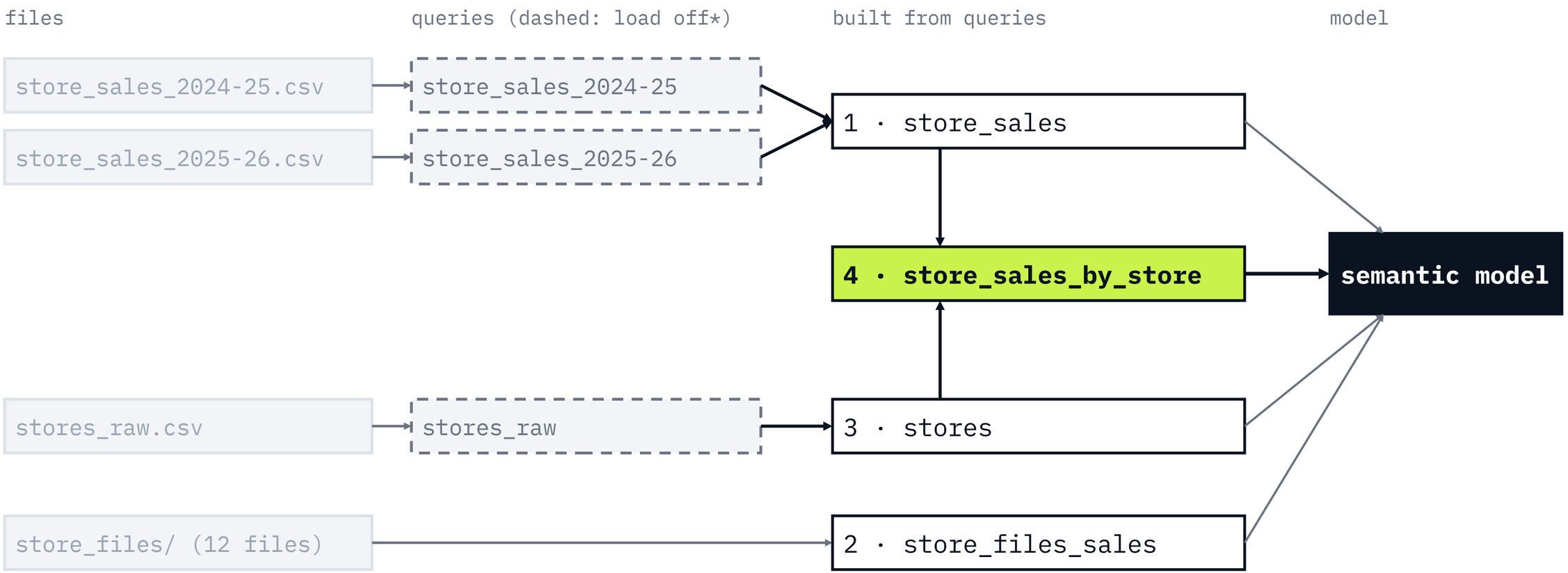
NK-POP-01, 18 completed sales: 858.00

Inner	revenue by governorate? 204,480.00: 858.00 short of the books	✗
Left Outer	revenue by governorate? 205,338.00, with a (Blank) bar to ask about	✓
Left Anti	sales with no store? 19 rows, all NK-POP-01 (18 completed)	✓
Right Anti	a shop with no sales? NK-JRS-01, Jerash: opens after the data ends	✓

For revenue by governorate, use Left Outer and say what the blank is.

THE QUERIES YOU BUILT TODAY · NUMBERED BY PART

Files in, one table per question out



The three sales routes each total 205,338.00; never add them together. Grey: comparison copies loaded today. *Where Enable load is available.

THE RULE TO KEEP

Read the estimate, then check the result.

After expanding, check the row count, unmatched rows and completed total. A matching total alone can hide the wrong groups.

STUDY LIST · IN CLASS AND IN THE LABS

Power Query today

In class	What it does	Activity
Append queries as new	rows of two queries in one table (UNION ALL)	01
Enable load	a helper query runs but is not loaded	01, 04
Folder → Combine Files	every file in a folder, with Source.Name	02
Reference	a query that starts from another query	04
Split Column → By Delimiter	one text column into several	04, 05
Keep / Replace Errors	find error cells; replace them with null	04
Column From Examples	a column from values you type	05
Merge queries as new	a join: Left Outer, Inner, the anti joins	06, 07

Lab	After class, self-guided, not submitted
08	**Desktop local paths: Manage Parameters; a status parameter**
09	**JSON → To Table, expand records and lists; one row per item**
10	**Remove Top Rows, Use First Row as Headers, errors that are sales**
11	**Unpivot Other Columns, Group By, a merge on two keys (optional)**

CLOSE

Checks for your report

- Append queries; names must match
- Combine a folder of files
- Split columns; fix error cells
- Merge; check rows, groups and totals

SAVE YOUR CHECKPOINT

D17-checkpoint.pbix, or browser report D17-checkpoint. Keep the data folder for refresh.